



The Nencki Affective Picture System (NAPS): a Children-Rated Subset

El sistema de imágenes afectivas de Nencki:
un subconjunto para niños

O sistema de imagens afetivas da Nencki:
um subconjunto de crianças

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Abstract

Emotion research in children should make use of emotional stimuli ideally coming from children-rated databases. However, most of the works in this area use adult-rated stimuli databases or a collection of stimuli taken from the Internet or from unidentified sources without any standardization procedure or content control for children population. These issues result in an overall lack of control over the kind of emotion to be elicited, leading to limited results. Therefore, in this work, a children-rated picture subset taken from the adult-rated Nencki Affective Picture System (NAPS) set is proposed as a stimuli database for emotion research in children. The NAPS is a standardized set of 1356 realistic, high-quality photographs classified into five categories (people, faces, animals, objects, and landscapes), and up to now, normative ratings were only collected in an adult population. The proposed subset in this work contains 1128 pictures taken from the original database which are developmentally appropriate for children based on different criteria and an expert's judgment procedure. Affective ratings were collected from a sample of 266 children 8–12 years old. With this work, both the availability and quality of affective stimuli for the study of emotion in children are directly improved.

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Resumen

La investigación en emociones en población infantil debería hacer uso de estímulos emocionales, idealmente provenientes de bases de datos calificadas por niños. Sin embargo, la mayoría de los trabajos en este campo utilizan bases de datos de estímulos puntuadas o validadas por adultos o estímulos tomados de Internet o de fuentes no identificadas sin ningún procedimiento de estandarización o control de contenido para población infantil. Estos problemas dan como resultado una falta general de control sobre el tipo de emoción que se debe provocar, lo que lleva a resultados limitados. Por lo tanto, en este trabajo, se propone un subconjunto de imágenes para niños del conjunto *Nencki Affective Picture System* (NAPS) desarrollado para población adulta. El NAPS es un conjunto estandarizado de 1356 fotografías realistas y de alta calidad clasificadas en cinco categorías (personas, caras, animales, objetos y paisajes) que hasta el momento solo cuenta con datos para población adulta. El subconjunto propuesto en este trabajo contiene 1128 imágenes tomadas de la base de datos original, apropiadas y aptas para población infantil, según los diferentes criterios y el procedimiento de juicio de un experto. Las evaluaciones afectivas se obtuvieron de una muestra de 266 niños de 8 a 12 años de edad. Con este trabajo, tanto la disponibilidad como la calidad de los estímulos afectivos para el estudio de la emoción en población infantil se ve incrementada.

Resumo

Pesquisas sobre emoções em crianças devem fazer uso de estímulos emocionais, idealmente a partir de bancos de dados avaliados por crianças. Entretanto, a maioria dos trabalhos nesse campo utiliza bancos de dados de estímulos pontuados ou validados por adultos ou estímulos retirados da Internet ou de fontes não identificadas, sem qualquer procedimento de padronização ou controle de conteúdo para crianças. Estes problemas resultam em uma falta geral de controle sobre o tipo de emoção a ser provocada, levando a resultados limitados. Portanto, neste trabalho, é proposto um subconjunto de imagens para crianças do *Nencki Affective Picture System* (NAPS) desenvolvido para a população adulta. O NAPS é um conjunto padronizado de 1.356 fotografias realistas de alta qualidade, classificadas em cinco categorias (pessoas, rostos, animais, objetos e paisagens) que até agora só têm dados para a população adulta e adulta. O subconjunto proposto neste trabalho contém 1128 imagens retiradas da base de dados original, adequadas e apropriadas para crianças, de acordo com diferentes critérios e um processo de julgamento de peritos. As avaliações afetivas foram obtidas a partir de uma amostra de 266 crianças de 8 a 12 anos de idade. Com este trabalho, tanto a disponibilidade quanto a qualidade dos estímulos afetivos para o estudo da emoção em crianças são aumentadas.

Keywords Affective visual stimuli · Affective rating · *Nencki Affective Picture System* (NAPS) · Children · Picture database

Palabras clave estímulos emocionales visuales · valoración afectiva · *Nencki Affective Picture System* (NAPS) · niños · base de datos de imágenes

Palavras-chave estímulos emocionais visuais · avaliação afetiva · *Nencki Affective Picture System* (NAPS) · crianças · banco de dados de imagens

Introduction

Nowadays, there are different theoretical models and definitions about emotion. Research in this field has shown that emotion is a process, which implies a set of behavioral, subjective, and physiological responses (LeDoux and Hofmann 2018). This phenomenon is supported by behavior studies and neuroimages, both of which use emotional-state induction methods with different strategies (Coan and Allen 2007; Dixon et al. 2017). Therefore, it is necessary to have emotional-state induction tools for the scientific study of emotions, with a theoretical foundation and adapted to the target population features and its context. Consequently, the development, update, and validation of emotional stimuli sets are of vital importance.

There are many standardized stimuli sets which are based both on the discrete and the dimensional model of emotions (Bujarski et al. 2015). The *discrete model* states that the emotional space is constituted by a set of specific but different affective states, such as fear, happiness, anger, disgust, sadness, and surprise (Ekman 1992). From this approach, it is assumed that each emotion has an appraisal pattern, an action tendency, a physiological response pattern, and a unique and distinguishable facial expression (Lench et al. 2011). However, this approach presents two important limitations associated to the number and identity of emotions. On the one hand, the number of basic emotions included in the theory vary as a function of the used paradigm and the type of emotion to be evaluated; thus, while some theories find six, others discriminate between seven, eight, and up to twelve basic emotions. On the other hand, the measures of the different basic emotions tend to be systematically interrelated. For example, sadness and anger could be grouped into a single measure of negative affect (Watson and Naragon-Gainey 2014). This situation hinders the measurement, assessment, and subsequent interpretation of discrete emotions. Therefore, the affective space could be better defined by a reduced number of general nonspecific dimensions upon which the concrete emotional states would be located, just like dimensional models propose (Mauss and Robinson 2009).

The *dimensional model* states that the affective space structure is supported by general continuous dimensions upon which concrete emotional states would be settled and defined (Lang, Greenwald, Bradley, & Hamm, 1993). The postulation of these dimensions is based on the seminal work by Osgood (1952) which introduces a multidimensional affective space with *valence* and *arousal* as its major variance components. Besides these two main dimensions, a dominance or control dimension was found but with less explanatory power. According to Bradley et al. (1993), this last dimension presents high correlations with the valence dimension and therefore adds little discrimination value. *Valence* means the way an individual assesses a stimulus quality (which ranges from pleasant to unpleasant). On the other hand, *arousal* reflects the activity generated by the stimulus on the person perceiving it (which ranges from excitement to calmness). Ultimately, the dimensional model has the advantage of defining the affective space by means of a reduced number of general continuous dimensions, and it lacks the characteristic overlapping of the discrete model (Lee et al. 2019). Despite being quite widespread, only a few emotional stimuli sets have been standardized for both the dimensional and the discrete models. Regardless of the approach to be used when studying emotion, stimuli can be obtained from standardized sets according to its auditory and verbal or visual modality.

Within visual stimuli sets, there are four standardized databases which contain general content, static affective visual stimuli: the International Affective Picture System (IAPS; Lang et al. 1999), the Geneva Affective Picture Database (GAPED; Dan-Glauser and Scherer 2011), the Emotional Picture System (EmoPicS; Wessa et al. 2010), and the Nencki Affective Picture System (NAPS; Marchewka et al. 2014).

From the previous sets, the IAPS is the most used and presents a large amount of validation studies. Moreover, it was validated in adult population within both the dimensional and the discrete models and presents standardized norms for children (e.g., Lang et al. 2008; Mina et al. 2017). In IAPS standardization studies, participants are asked to see images and rate each of them in arousal and valence scores, using a pictographic Likert scale known as Self-Assessment Manikin (SAM; Bradley and Lang 1994). Thus, valence and arousal mean scores are rendered in a bidimensional map which represents the affective space. In general, a characteristic boomerang-shaped pattern has been found, which reveals that negative valence images show greater arousal and that neutral valence is associated with low arousal levels. Despite being quite widespread, the IAPS dataset suffers from certain shortcomings: (a) it lacks explicit semantic categories, which complicates its experimental usage when choosing stimuli by content; (b) image quality is outdated for the latest display screens; (c) even though there are stimuli for children, these amount for a reduced subset (60 images), which represents an issue for studies requiring a large amount of stimuli (e.g., neuroscience studies); (d) the picture format is inhomogeneous, i.e., it varies in resolution, size, and shape along the whole set, which constitutes a big disadvantage because different studies have shown that the physical properties of an image, such as size, luminance, and complexity can affect visual stimuli processing (Marchewka et al. 2014; Rozovskaya et al. 2016).

The GAPED (Dan-Glauser and Scherer 2011) is a multiple-categories database, where negative pictures categories are spiders, snakes, or moral and/or legal norms violation scenes. Positive pictures categories are babies and animal pups, while neutral pictures correspond to objects. Despite its specific content, its main limitations are small picture size and asymmetrical content (mostly unpleasant stimuli). The EmoPicS database (Wessa et al. 2010) was developed as a supplement to the IAPS and adds luminance, contrast, and color composition to each picture of the set. However, it contains a moderate amount of stimuli, including 378 pictures selected from the Internet and various image archives, with variable resolution.

Last but not least, the NAPS is a database containing 1356 realistic pictures divided into five categories (people, faces, animals, objects, and landscapes), with controlled luminance, contrast, and entropy, which ease its selection, manipulation, and later evaluation of the obtained results and give an internal validity warranty (Grün & Sharifian, 2016). Also, this database has norms within both the dimensional (Marchewka et al. 2014) and the discrete (Riegel et al. 2016) models. Recently, there has been an attempt to compare the NAPS affective ratings between Iranian and European population (Riegel et al. 2017), but no differences were found between both models' ratings taking culture as its variation factor, in spite of previous studies which hypothesized the contrary (Hampton and Varnum 2016). Likewise, it has been found that affective space distribution proposed by the NAPS shows that the valence-arousal relationship can be described by a quadratic function which has a boomerang shape (Riegel et al. 2016, 2017) or by a linear function (Marchewka et al. 2014).

When using emotionally charged material in experimental tasks, it is possible to consider age-related emotional elicitation differences. A pioneer study of age-related score differences was presented by McManis et al. (2001) who reported that older children and adults showed similar valence and arousal patterns using the IAPS. However, younger children rated unpleasant images less arousing and pleasant or neutral images more arousing than older children and adults did. In concordance with this result, an IAPS stimuli subset was developed (Lang et al. 2008) in order to make norms for three children age groups (7–9, 10–12, and 13–14 years old). At the same time, these ratings varied between children age groups. Nevertheless, the children IAPS subset includes content not suitable for their age, raising a debate of its usefulness (Cordon et al. 2013; Zamora et al. 2017). For this reason, a visual stimuli database called Developmental Affective Photo System (DAPS) was established for children 7–9 years old only (Cordon et al. 2013) and features valence, arousal, and complexity scores. This database contains 593 images from various sources (including 136 IAPS images, the rest taken from books, magazines, and the Internet) and showing different kinds of scenes (e.g., objects, landscapes, faces, accidents, weapons). Unfortunately, to our knowledge, the DAPS is not available for its use in developmental research experiments.

As a brief summary, the main issues present in developmental research works using emotional stimuli are (a) the use of standardized adult-rated stimuli with children and (b) the use of images or photographs taken from the Internet or from unidentified sources without any standardization or content control for children population. These issues result in an overall lack of control over the type of emotion to be elicited in emotion research with children.

Current Study

The previously mentioned issues explain the lack of available affective stimuli both suitable for and rated by children. Therefore, the main objectives of this study were to obtain a large set of visual stimuli suitable for children and to add children affective ratings to each stimulus, in order to have a children-rated affective stimuli set for developmental research. The emotional stimuli subset was obtained through a three-stage process: (1) selection of stimuli appropriate for children, (2) selection of a Likert scale for the scoring procedure with a graphic format appropriate for children, and (3) group slideshows for evaluation of each selected photograph. Likewise, the scores between the different children and adult age groups will be compared in order to analyze their differences.

Material and Methods

Participants

The sample consisted of 266 (141 girls and 125 boys) children with ages between 8 and 12 years old ($M = 10.08$ years; $SD = 1.07$), receiving elementary education at one of two selected private schools in residential areas of Mar del Plata, Buenos Aires, Argentina. It was obtained via a convenience sampling process and split into two

groups: a younger children group ($G1 = 145$) comprised of children in third and fourth grade with an age mean of 9.36 years old ($SD = 0.53$) and another group of older children ($G2 = 121$) comprised of children in fifth and sixth grade with an age mean of 11.21 years old ($SD = 0.61$). The conformation of the groups is defined according to the Technical Report A-8 (Lang et al. 2008), in which the IAPS instruction manual and normative scores are presented for US children.

In order to inquire the socio-educational level of the children families, a semi-structured survey was sent to the children's parents or legal representatives which gathers educational and occupational level of the family. It is worth noting that only the 51.5% of the families ($n = 137$) completed the survey. Of the complete surveys, 52.5% of them were completed by the mother of the child and 34.3% by the father, while the rest of them were completed by other parents such as the grandmother or the grandfather. Regarding educational level, data showed that a 13% of the parents or tutors had university degrees, 20% had unfinished university degrees, 25% had completed secondary education, 18% had unfinished secondary studies, 9% had completed elementary education, and 3% had unfinished elementary studies. Regarding occupational level, 28.5% were qualified or semi-qualified workers (electricians, turners, carpenters, drivers, low-ranked soldiers), 17.5% were unqualified workers (waiters, masons, laborers, assistants, servants or maids), 17% were qualified managers, 7% were salesmen or saleswomen, 9% were technicians or small business owners, 4% were commercial store managers or owners, and 3% were professionals or senior executives. For its classification, the Scale of Occupational Groups of Argentina EGO 70 (Sautú 1989) was used. The data obtained are representative of the Argentine population, according to the data reported by the Argentine Social Debt Observatory, UCA (2010–2018).

All participants were children with normal or corrected vision, not under psychiatric treatment at the time of the evaluation and without a history of learning or neurodevelopmental problems according to a health record completed by their parents. For the analysis of this study, 2 children were excluded from the sample because they were in school integration.

Ethical Considerations

This study was carried out in accordance with the recommendations of *The Programa Temático Interdisciplinario en Bioética* (PTIB, The Interdisciplinary Thematic Program in Bioethics) of the Universidad Nacional de Mar del Plata (UNMDP, National University of Mar del Plata). The PTIB approved the protocol and the final set of pictures used in this study. An informed consent was required to be read and signed by parents or legal guardians, which contained the objectives and procedures of the study. It was explained that participation was voluntary and anonymous and also that the child could quit the study voluntarily. Then, children were informed about the study, and only the ones who gave their assent to participate in the study. All subjects gave written informed consent in accordance with the Declaration of Helsinki.

Material

The emotional stimuli subset was obtained by means of a three-stage process:

1. Selection of suitable stimuli for children through empirical criteria found on the literature, concordance between expert's judgment, and ethical aspects assessment of the selected content
2. Selection of an appropriate Likert scale for the rating procedure with children
3. Stimuli presentation in groups for the collection of valence and arousal ratings for each picture

Each of these stages will be explained in the following subsections.

Selection of Suitable Stimuli for Children

First, as seen in Table 1, step a, an image selection process was carried out considering as its main inclusion criterium their adequacy to the children development stage, based on previous studies (e.g., Brenner, 2000; Coan and Allen 2007; Hajcak and Dennis 2009) and regulations that led to the exclusion of some images which showed nudity and death (shown both explicitly or implicitly, of humans or animals). In the absence of an explicit stimuli selection methodology for the different age groups including ethical considerations in children population, an external reference providing a normative for its regulation and selection was sought. In Argentina, the *Ley de Servicios de Comunicación Audiovisual* ([Audiovisual Communication Services Law]; 2009) regulates suitable content for everyone, excluding nudity, blood, and alcohol or drug use. Additionally, the *Comisión Asesora de Exhibición Cinematográfica* (CAEC [Cinematographic Exhibition Advisory Commission]) dispositions were taken into account, which by complying with Ley 23.052 about film movies qualification regime state that children shall be protected against exhibitions which imply a certain danger of intellectual, affective, or moral disturbance. Taking both criteria into account, 204 NAPS photographs were excluded¹ for showing the following content: nudity, explicit or implicit violence, weapons, blood, maiming (either in animals or in human beings), death (explicit and implicit content both in humans and animals), terminal illness references, and concentration camp images.

In second place (Table 1, step b), an expert judge panel was established with the objective of determining if the selected set was suitable for children ages 8–12 years old. Argentinian and Brazilian professional experts and specialists with extensive experience in psychological assessment, emotion, and developmental psychology were selected. They were asked to fill assessment questionnaires and given an online link where all images were available for the judges to see them. With the objective of easing the assessment process, 6 groups of 2 experts each were created, assessing 192

¹ The following NAPS photographs were excluded: Animals (1, 8, 10, 13, 16, 17, 18, 19, 24, 25, 27, 29, 32, 33, 38, 39, 45, 47, 48, 49, 54, 56, 57, 60, 62, 63, 64, 67, 68, 71, 73, 74, 75, 77, 78, 450, 204, 207), faces (3, 7, 9, 10, 13, 16, 18, 19, 25, 27, 28, 29, 31, 32, 33, 34, 38, 52, 104, 141, 143, 145, 147, 149, 151, 153, 159, 160, 170, 174, 264, 265, 266, 268, 269, 270, 271, 272, 273, 274, 275, 280, 284, 291, 293, 298, 300, 330, 345, 362, 363, 364, 365, 366, 367, 368, 369, 371), landscapes (1, 2), objects (1, 5, 6, 7, 12, 13, 15, 17, 18, 19, 21, 23, 24, 55, 87, 98, 103, 105, 115, 118, 119, 121, 126, 127, 131, 132, 137, 139, 143, 145, 147, 148, 149, 154, 328, 285), and people (2, 3, 7, 10, 13, 19, 20, 22, 23, 24, 25, 31, 34, 36, 37, 38, 39, 40, 42, 53, 57, 58, 59, 60, 62, 64, 70, 71, 72, 73, 74, 75, 76, 77, 79, 84, 86, 88, 124, 127, 128, 129, 140, 147, 155, 181, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209, 211, 212, 213, 214, 215, 216, 217, 218, 220, 221, 222, 223, 224, 225, 226, 227, 228, 229, 230, 231, 232, 233, 234, 235, 236, 237, 238, 240, 241, 242, 243, 245, 246, 247).

Table 1 An overview of the picture subset selection procedure

Stages (Initial set = 1356 pictures)	Amount of excluded images
Selection	204 pictures excluded for noncompliance with criteria taken from previous studies and regulations.
Expert assessment	18 pictures excluded according to the judgment of experts in child development, emotion and evaluation
Ethical committee assessment	0 pictures excluded
Excluded by error*	6 pictures
Resulting NAPS subset	Final set = 1128 pictures

*Unintentional mistakes when in set building

photographs per group. Images belonging to each assessment group were randomized without affecting the amount per category of the original database.

Then, a concordance analysis between experts was performed; for this, the kappa index, κ , recommended for qualitative variables—nominal in this case—was used (Carletta 1996). As a result, another 18 images were excluded based on a strong-almost perfect concordance for its removal (see Table 2). As a result of this procedure, the final 1134 photographs were sent to the PTIB for its assessment, with favorable results for its usage in children population.

Selection of an Appropriate Likert Scale

The Self-Assessment Manikin (SAM) (Bradley and Lang 1994) is a Likert scale which was used for collecting affective ratings for the IAPS. However, given the scarcity of studies with children and the discrepancy between the different response options² in the SAM, a SAM response level section was also included for the experts to assess the optimal amount of scale points (3, 5, 7, or 9) for the photographs' valence and arousal ratings which would be given by children. Of these, 58% of the experts ($n = 7$) opted for the option with 5 response points for valence and arousal. Seventeen percent of the experts chose the option with 3 response points, another 17% with 7 response points, and only one expert chose the option with 9 points. Based on these results, the 5-item SAM scale was selected for its use in the picture rating process (Fig. 1).

Rating Process

The task was administered in groups on a standard projector in a quiet place and lasted approximately 45 min. The maximum size of each picture projected was standardized at approximately 120 cm × 150 cm. Images were rated by 266 participants split in groups of 15 participants each. Group interviews were made at school time and were located in a classroom lent by the institution. By means of using the SAM scale in pen and paper as a 5-point Likert scale as advised by the experts, children marked the

² For example, Hajcak and Dennis (2009) used 9 response points; Cordon and cols. (2013) used 5 response points.

Table 2 Degree of concordance between experts

Image group	κ	Degree of concordance*
1	0.57	Moderate
2	0.50	Moderate
3	1	Almost Perfect
4	0.70	Strong
5	1	Almost Perfect
6	0.65	Strong

κ ($p < 0.05$); *Based on Landis and Koch (1977)

valence and arousal scores for each of the pictures displayed using a video projector. Rating sessions lasted for approximately 45 min, showing 95 different images per session. Before starting the experiment, it was explained to the children that they would see pictures showing different things which could make them feel happy or sad, active or relaxed, or maybe angry or afraid and that each child could feel different about each picture. The rating procedure was also carefully explained to them, its objective was stated, and different practice trials were made. The guidelines of the study are found in [Supplementary Material N°2](#).

Approximately 95 stimuli were shown to each participant selected randomly from the 5 categories. A total of 12 stimuli sets were prepared from the 6 picture groups. On

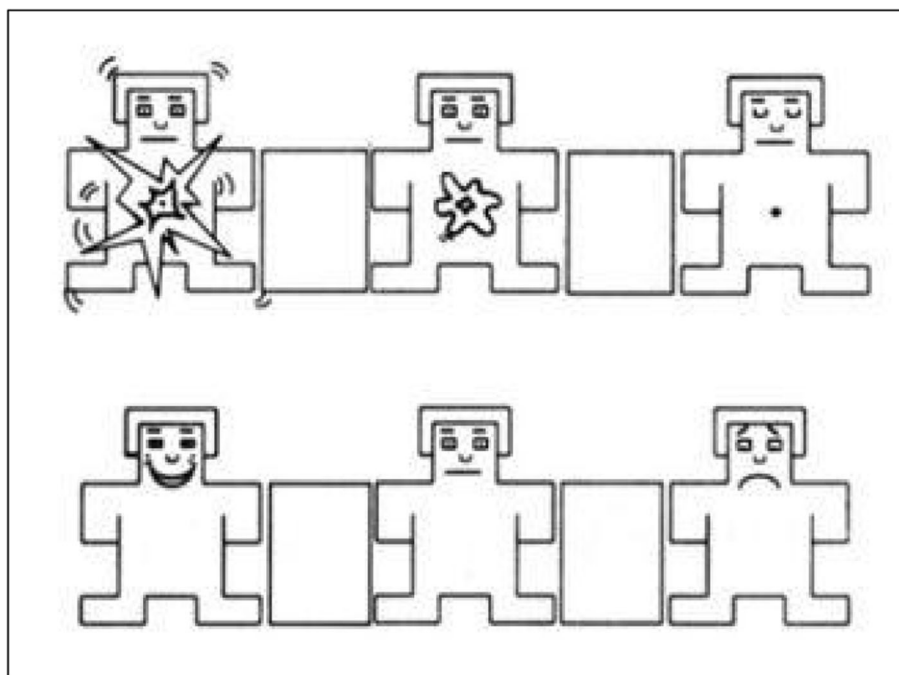


Fig. 1 Pen and paper 5-point Likert version of the SAM Scale used in this study. Arousal and valence dimensions are shown in the upper and lower row, respectively

average, approximately 25 ratings were collected to each picture. The sessions started by reading the instructions and doing 8 practice trials. Then, each trial began with a preparation slide (“Look at the sheet and line number 1. There you must answer, please be ready”) which was shown for 5 s. After that, a fixation cross appeared for 500 ms, before the picture to be rated was shown on screen for 6 s, and after disappearing, the SAM scale was shown on screen for a standard period of 15 s indicating that valence and arousal scores should be given to the picture shown a moment before (Fig. 2).

Results

Analysis of Age-Related Differences in Children

Affective ratings for each picture, as well as their description, are presented in the supplementary materials ([Supplementary Material N°1](#)). The descriptive statistics for the different dimensions and group categories are shown in Table 3.

As can be seen in Fig. 3, a monotonic relationship between valence and arousal exists for each category and age group. Therefore, in order to analyze possible emotional response differences, we first calculated Pearson’s correlation for each age group and category as shown in Table 4, and then we used the test of the difference between two independent correlation coefficients (Preacher 2002), the correlation coefficients between groups. This procedure gives information about how different are the emotional reactions between two groups for the same category of images. This calculation requires the conversion of both correlation coefficients into a z score, which can be done using the Fisher’s r -to- z transformation. By this procedure, valence and arousal dimensions in children only had significant correlation differences in the objects category ($z = 4.673$, $p < 0.001$, effect size = 0.39). This suggests that different emotional responses can be elicited by the same pictures in the objects category as a function of age.

Comparison with Adults

In order to be able to compare children ratings obtained with the 5-point Likert scale with adults ratings obtained with its 9-point counterpart (Riegel et al. 2016), the adults’ SAM ratings given to part of the NAPS subset ($N = 395$)³ for children were transformed using the following function:

$$TS = 0.5 OS + 0.5,$$

where TS is the transformed score and OS is the original score. For example, if OS was 1, then TS results in 1, but if OS was 9, then TS results in 5. By using this transformation, the original rating is transformed to an equivalent 5-point Likert scale

³ The standards available for adults were used, and they were not administered in this work. The adult-rated NAPS using the SAM scale is comprised of 510 pictures of which 395 coincide with the 1128 pictures of the proposed NAPS subset for children.



Fig. 2 Picture slideshow sequence

rating, without losing information due to the fact that both intervals are dense sets, i.e., contain real numbers. The descriptive statistics for the different dimensions and group categories are shown in Table 5.

As can be seen in Fig. 4, even though the children's affective space shows a monotonic relationship between valence and arousal, this is not the case for the adults'

Table 3 Descriptive statistics, calculated separately for each dimension and category

		Younger group valence	Younger group arousal	Older group valence	Older group arousal	Both groups: valence	Both groups: arousal
Faces	Mean	2.87	2.28	2.93	2.00	2.89	2.15
	SD	0.82	0.50	0.75	0.58	0.70	0.44
	Min-Max	1.20–5.00	1–3.90	1.2–4.90	1–4.20	1.37–4.81	1.26–3.53
	<i>n</i>	313	313	313	313	313	313
People	Mean	3.24	2.53	3.04	2.36	3.15	2.45
	SD	0.94	0.63	0.97	0.60	0.90	0.50
	Min-Max	1.2–5.00	1.17–4	1.10–5.00	1.20–4.10	1.20–4.93	1.41–3.74
	<i>n</i>	157	157	157	157	157	157
Animals	Mean	3.75	3.02	3.65	2.94	3.71	2.99
	SD	0.98	0.5	0.89	0.69	0.89	0.48
	Min-max	1.40–5.00	1.73–4.50	1.20–5.00	1.50–4.80	1.30–5	1.90–4.55
	<i>n</i>	183	183	183	183	183	183
Objects	Mean	3.11	2.53	2.97	2.26	3.05	2.41
	SD	0.91	0.61	0.83	0.54	0.80	0.47
	Min-max	1.00–5.00	1–4.40	1.20–5.00	1.20–4.70	1.40–4.89	1.40
	<i>n</i>	292	292	292	292	292	292
Landscapes	Mean	3.68	2.69	3.50	2.34	3.60	2.53
	SD	0.84	0.57	0.80	0.63	0.78	0.48
	Min-max	1.50–5.00	1.33–4	1.5–5.00	1.10–4.30	1.80–4.77	1.50–3.65
	<i>n</i>	183	183	183	183	183	183
General	Mean	3.26	2.57	3.17	2.33	3.22	2.46
	SD	0.95	0.61	0.89	0.67	0.86	0.54
	Min-max	1.00–5.00	1.00–4.50	1.10–5.00	1.00–4.80	1.20–5.00	1.26–4.55
	<i>N</i>	1128	1128	1128	1128	1128	1128

The *n* under each category refers to the number of stimuli

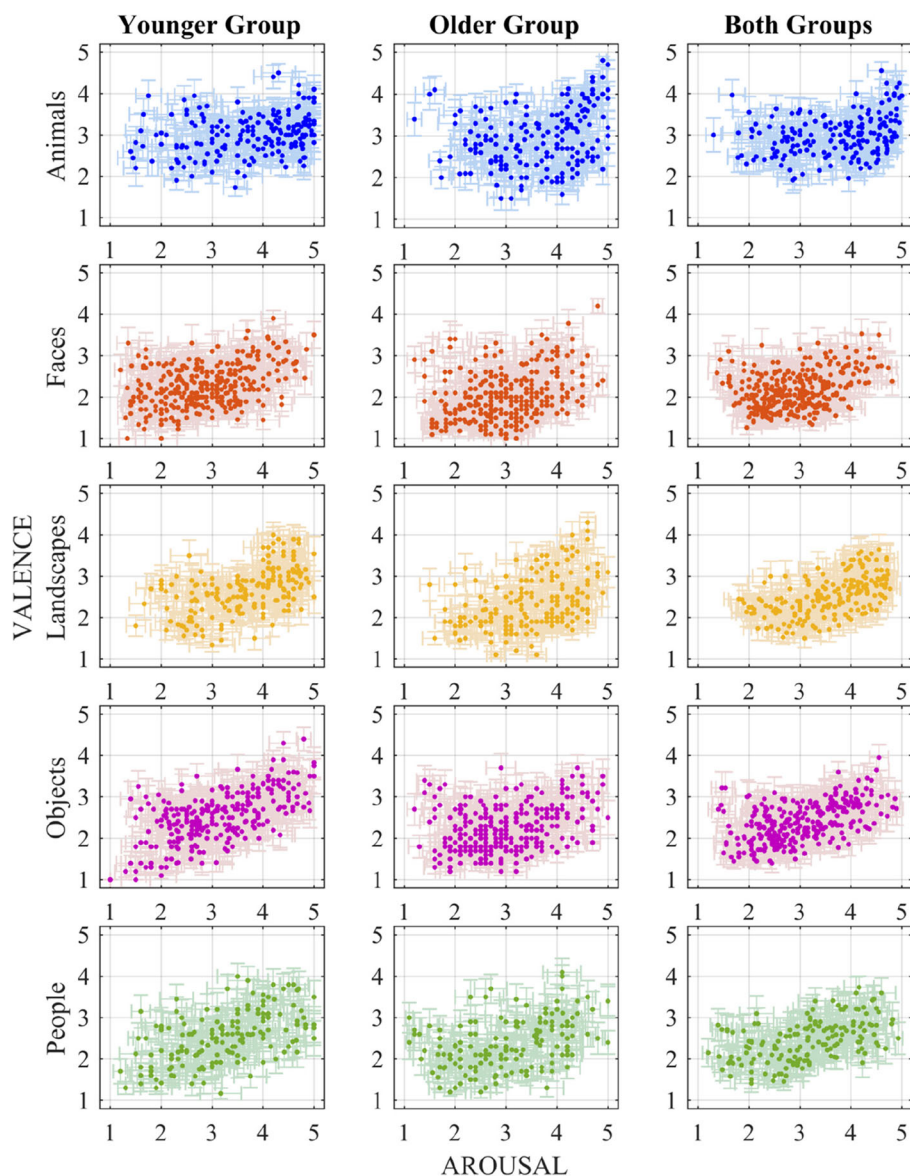


Fig. 3 Ratings for valence (y-axis) and arousal (x-axis) in each category for the children younger group, older group, and both groups. Each single dot represents the mean rating for a particular image on a two-dimensional scale, with rating standard deviations (SDs) in light colors

affective space, where a non-monotonic *U*-shaped relationship is seen. Therefore, a correlation analysis in this case is not the most appropriate means for comparing both affective spaces. In this case, a curve fitting analysis was made in order to extract a possible model for the valence-arousal relationship.

In the case of children, the best fit was found with a linear relationship [$F(1,393) = 127.641, p < 0.001$] with $R^2 = 0.245$. Arousal scores (y-axis) can be predicted with

Table 4 Correlation coefficients of valence and arousal ratings, listed for each category and group

Category	Dimension	Younger group Arousal	Older group Arousal	Children (both groups) Arousal
Faces	Valence	0.397**	0.325**	0.349**
People	Valence	0.534**	0.383**	0.495**
Animals	Valence	0.271**	0.234**	0.288**
Objects	Valence	0.639**	0.352**	0.524**
Landscapes	Valence	0.525**	0.438**	0.559**

** $p < 0.001$

valence scores (x -axis), by the following equation: $y = 1.479 + 0.305x$. This means that for each valence score unit increment, arousal is incremented by 0.305 units. On the other hand, in the case of adults, the best fit was found with a cubic relationship [$F(2, 392) = 202.667$, $p < 0.001$] with $R^2 = 0.508$. Arousal scores (y -axis) can be predicted with valence scores (x -axis), by the following equation: $y = 6.345 - 2.209x + 0.074x^3$.

Table 5 Descriptive statistics, calculated separately for each dimension in children and adults for a NAPS subset (Riegel et al. 2016)

		Children: valence	Children: arousal	Adults: valence	Adults: arousal
Faces	Mean	2.94	2.18	3.24	1.89
	SD	0.75	0.43	0.57	0.32
	Min-max	1.36–4.81	1.39–3.5	1.52–4.13	1.24–3.10
	n	135	135	135	135
People	Mean	3.14	2.45	3.14	2.03
	SD	0.95	0.49	0.71	0.41
	Min-max	1.20–4.81	1.40–3.60	1.60–4.23	1.28–3.37
	n	51	51	51	51
Animals	Mean	3.73	2.98	3.26	2.11
	SD	0.89	0.47	0.51	0.32
	Min-max	1.30–4.95	1.94–4.25	2.03–4.41	1.46–2.87
	n	76	76	76	76
Objects	Mean	2.99	2.37	2.98	1.69
	SD	0.79	0.47	0.41	0.32
	Min-max	1.46–4.89	1.40–3.40	1.74–4.10	1.24–2.92
	n	85	85	85	85
Landscapes	Mean	3.74	2.57	3.39	1.91
	SD	0.81	0.54	0.64	0.39
	Min-max	1.95–4.72	1.50–3.60	1.78–4.37	1.31–2.69
	n	48	48	48	48

The n under each category refers to the number of stimuli

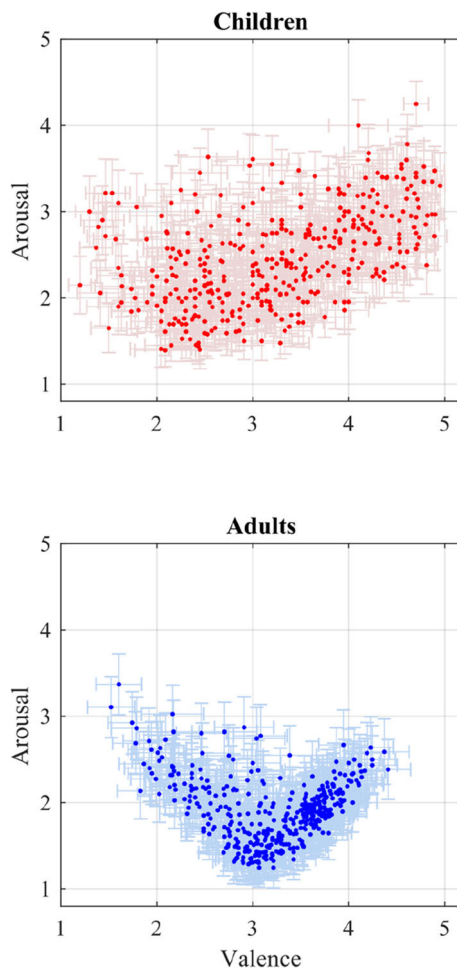


Fig. 4 Affective space according to valence and arousal dimensions for children and adults corresponding to the proposed NAPS subset

Discussion

Children emotion research should make use of emotional stimuli ideally from children-rated databases. However, most of the works in this area use adult-rated stimuli databases or a stimuli collection taken from the Internet or from unknown sources without any standardization procedure or content control for children population. Moreover, there is evidence that the same stimulus could elicit different emotional reactions in children than in adults. These issues result in a global lack of control over the type of emotion to be elicited, leading to limited results. Because the NAPS is a relatively large, varied, and high-quality stimuli set, it could be of interest for behavior and neuroscience research. Therefore, this study had the objective of obtaining a children-rated NAPS subset, which was previously rated by adults, but by adding children scores to this subset, now suitable for research with children population.

By comparing the 8–10 years old (younger group) with the 10–12 years old (older group) children ratings, results showed that the children affective ratings do not present significant age-related differences, except in the objects category where there is a small valence-arousal correlation difference. This means that the proposed children-rated NAPS subset could be used in children population in almost all of its categories, excluding the objects category which presents differences between younger and older children.

Due to the fact that children affective ratings differ from adult affective ratings for the same stimulus (Cordon et al. 2013), significant differences were expected to be found between the original adult ratings and the newly acquired children ratings. Our results show that children affective ratings were best described by a linear function, showing a positive valence-arousal relation, in contrast with the adults' ratings which were best described by a cubic function. The linear valence-arousal relationship found in children can be possibly due to the lack of exciting negative stimuli, e.g., dead humans or animals. However, it is important to note that the affective dimensional space found in this work is in line with previous results obtained in the original study with adult population (Marchewka et al. 2014). These results show that using the adult-rated NAPS with children could lead to a different from the expected emotional reaction, highlighting the importance of having one or more children-rated stimuli subsets for its use with children population and the need to obtain children ratings for existing adult-rated affective stimuli databases.

Due to the fact that the NAPS pictures were subjected to a selection process in order for them to be suitable for children population, extremely unpleasant pictures were excluded, and therefore, the affective space of the proposed NAPS subset lacks from extremely negative and highly arousing pictures. On the contrary, the children IAPS affective space (McManis et al. 2001; Lang et al. 2008) has a boomerang shaped plot, showing the absence of extreme valence (positive or negative) low arousing pictures, even though using extremely negative IAPS pictures with children is questionable. However, the children-rated NAPS subset contains negative-valence low-arousal pictures, supplying stimuli for this part of the affective space.

Notwithstanding the evidence provided in this work, our results should not be taken lightly, and further works should be made to support our findings. First of all, it is important to note that the ratings were collected from 8 to 12 years old children, and they can differ from the ratings of other populations (e.g., clinical samples, teenagers, or younger children). As an example, it is important to consider the possibility of eliciting a different emotional reaction in clinic population when using aversive pictures of the proposed NAPS subset. Also, it would be important to analyze the moderating roles of gender and socioeconomic status, as they have shown to exert an effect on cognitive variables. Finally, another aspect which should be analyzed is the effect of the scale used to collect affective picture ratings. In order to compare adults' ratings with the children's ratings, we transformed the original adults' ratings, which were collected using a 9-point Likert scale, to a range between 1 and 5, which corresponded with the 5-point Likert scale used with children. However, this transformation does not capture any possible bias between using a 5- or 9-point Likert scale with adults. Therefore, further studies should be made to support the results provided in this work.

Despite these limitations, a children-rated NAPS subset suitable for children was developed, which represents an attractive alternative to other known databases, such as

the IAPS, which have led to controversial research results, as in fMRI studies (Kandilarova et al. 2018). To this end, a 1128 NAPS pictures subset was selected and later rated in valence and arousal dimensions by children 8–12 years old. This is an essential contribution which opens the perspective for the development of a psychological and neuroscientific evaluation instrument for children based on a visual affective system, beyond the existing systems of the same type applied in adult populations.

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